

Gabriel Rabanal

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WORK EXPERIENCE

4G Clinical

Software Engineer

Wellesley, MA

Sep 2024 — Present

- Developed Python automation and reusable validation scripts to implement complex study logic, prevent configuration errors, and ensure reproducible configurations
- Developed and maintained production configurations for clinical study platforms, ensuring validated data workflows and reporting in a regulated (GxP) environment
- Collaborated with engineering and clients to troubleshoot issues and ensure reliable system behavior and audit-ready configurations

Harvard University and CERN

Doctoral Researcher (ML & large-scale data analysis)

Cambridge, MA / Geneva, Switzerland

Sep 2018 — Nov 2023

- Developed and validated ML classifiers (DNNs, BDTs) on multi-terabyte particle physics datasets using HPC batch computing; improved signal detection significance by $+3\sigma$
- Built automated data-quality and model-validation pipelines for large-scale analyses to reduce performance regressions and improve reliability (statistical tests, reproducibility checks, uncertainty quantification)
- Commissioned and calibrated detector subsystems to improve tracking efficiency and spatial resolution; mentored researchers and coordinated cross-institution deliverables

Yale University

Lab Researcher

New Haven, CT

Jan 2016 — Apr 2016

- Analyzed detector data and built Monte Carlo simulations for the PROSPECT neutrino experiment

Peruvian Institute of Nuclear Energy

Undergraduate Researcher

Lima, Peru

Jan 2015 — Apr 2015

- Calibrated spectrometers for neutron flux measurements in a research nuclear reactor supporting medical isotope production

EDUCATION

Harvard University

Ph.D. in Physics

Cambridge, MA

Nov 2023

National University of Engineering

B.S. in Physics, GPA: 4.00/4.00

Lima, Peru

Jan 2016

SKILLS

Programming: Python (advanced), C/C++ (advanced), SQL (working knowledge)

Machine Learning: TensorFlow/Keras, XGBoost, scikit-learn; feature engineering, model validation

Data & Statistics: NumPy, Pandas, SciPy; Bayesian and frequentist methods; data pipelines

Tools: Git, Linux/Bash, Jupyter, VS Code; testing, debugging, HPC environments

Languages: Spanish (native), English (fluent), French (advanced)

Other: Linguistics; Akkadian, Sumerian, Quechua, Latin

SELECTED PUBLICATIONS

ATLAS Collaboration. [Observation of \$VVZ\$ production at \$\sqrt{s} = 13\$ TeV](#) (Phys. Lett. B, 2025) [[arXiv](#)]

ATLAS Collaboration. [Evidence for three massive vector bosons](#) (Phys. Lett. B, 2019) [[arXiv](#)]

Rabanal Bolaños, G. [Micromegas sectors for the ATLAS muon upgrade](#) (PoS ICHEP2020, 2021)